

JOSE VILLEGAS

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EDUCATION

Michigan State University

East Lansing, MI

Bachelor of Science in Computational Data Science

GPA : 3.4/4.00 | May 2026

Relevant Coursework: Big Data Analysis, Database Systems, Fundamentals of Data Science Methods, Probability and Statistics for Data Science, Introduction to Machine Learning, Introduction to AI, Data Structures and Algorithms, Matrix Algebra, Optimization Methods in Data Science, Numerical Analysis, Introduction to Parallel Computing

WORK EXPERIENCE

Delta Dental

Okemos, MI

Data Engineer Intern

January 2026 - April 2026

- Designed a canonical data model across 4 tables with over 20+ fields, transforming claims data into a scalable analytics pipeline
- Developed a Python package and 15+ FastAPI endpoints, streamlining data processing and querying for downstream analytics
- Presented technical trade-offs to 10+ stakeholders, securing buy-in by aligning system design with business needs

Autonomous Vehicle Club

East Lansing, MI

Software Engineer

September 2024 - December 2025

- Tested 3 autonomous driving algorithms across 6 simulated F1 tracks to evaluate navigation and minimize wall collisions
- Optimized navigation algorithms, reducing wall collisions from 10+ to fewer than 10 per simulation and improving consistency
- Collaborated with 2 club members to evaluate 3 navigation algorithms across 6 tracks and refine autonomous driving strategies

Wayne County Prosecutors Office

Detroit, MI

IT Intern

July 2024 - August 2024

- Cut new-intern onboarding time by 33% by implementing a Jira workflow map, streamlining the process for incoming interns
- Reduced recurring pipeline errors by building custom Jira issue types, improving ticket routing and resolution across the office
- Produced an improved intern onboarding process to ensure more efficient and consistent workflows for employees and interns

PROJECTS

Automatic Stock Analyzer | Python, PostgreSQL, SQL, Power BI

July 2026

- Built an ETL pipeline for 3 months of daily stock data using Python and Alpha Vantage, loading 500+ records into PostgreSQL
- Implemented incremental data-loading logic to track the latest trade date per ticker and preventing redundant database writes
- Developed a Power BI dashboard connected to PostgreSQL to visualize closing-price and trading trends across 5 companies

Chest X-Ray Disease Detection | Python, TensorFlow, PyTorch

April 2025

- Designed a deep learning pipeline with TensorFlow and DenseNet121 for chest X-ray classification on 3 different diseases
- Streamlined data preprocessing and caching for large-scale medical imaging data, boosting training and validation speed by 30%
- Tuned convolutional neural network hyperparameters to achieve 0.9495 binary accuracy and 0.857 multi-label accuracy

House Price Predictor | Python, Pandas, Scikit-learn, NumPy

March 2025

- Engineered predictive features on a dataset of 200K+ records, reducing dimensionality by eliminating low-impact variables
- Benchmarked and optimized multiple regression models, achieving an R^2 score of 0.974 on held-out validation data
- Deployed a house price prediction web application leveraging 10+ features to provide real-time property price inference

TECHNICAL SKILLS

Languages: Python, SQL, R, MATLAB, Bash/Shell Scripting, C, C++

Data Engineering & Databases: PostgreSQL, Snowflake, Database Systems, Big Data Analysis, Docker

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Statsmodels.api, FastAPI, Flask, Keras

Tools: Git, Jira Software, Power BI, Foxglove, OpenMP, MPI, Matplotlib, Plotly